Coding & Solution

Date	30 September 2026
Team ID	SWTID-2026-4536
Project Name	LegalEase – AI-Powered Legal Document Generator
Maximum Marks	5 Marks

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LegalEase is an AI-powered legal document generation system designed to simplify the creation of structured legal documents. The application uses Generative AI to create customizable and editable legal documents based on user-provided information such as document type, parties, terms and effective dates.

The solution uses **Python, FastAPI, Streamlit and Google Gemini API**. The system supports legal document generation, editable preview, and document export in **TXT, DOCX and PDF** formats.

The backend provides a /generate API endpoint that accepts document details and sends them to the AI generation module. The frontend provides an easy-to-use Streamlit interface for entering information, generating documents, editing the generated content and downloading the final document.

Instructions

1. The code is structured using separate frontend, backend, AI integration and document formatting modules.
2. The solution implements the core requirements of AI-powered legal document generation.
3. Required configuration and dependency information is provided through the project setup and requirements.txt.
4. The project can be connected to a GitHub/GitLab repository for version control.

Solution Summary

S.No	Field	Details

1	**Repository Link / URL**	[Paste your GitHub / GitLab or hosted link here]
2	**Programming Language(s)**	Python
3	**Framework(s) Used**	FastAPI, Streamlit
4	**AI Technology**	Google Gemini API
5	**Key Features Implemented**	AI-powered legal document generation, editable document preview, document formatting, TXT/DOCX/PDF export, user input handling, AI integration
6	**Pending / Incomplete Features**	GitHub repository link and cloud deployment can be added when available
7	**Setup / Run Instructions**	Install Python dependencies, configure the Gemini API key in .env, start the FastAPI backend using Uvicorn, and run the Streamlit frontend

Key Features Implemented

1. **Legal Document Generation**
    - Generates legal documents based on document type, parties, terms and effective dates.
    - Supports documents such as contracts, NDAs, lease agreements and employment documents.
2. **AI Integration**
    - Integrates Google Gemini for AI-powered legal document generation.
    - The GeminiDocumentGenerator creates structured prompts and sends them to the AI model.

3. **Backend API**
    - FastAPI is used as the backend framework.
    - The /generate endpoint processes document-generation requests.
    - Pydantic is used for request validation.
4. **Frontend**
    - Streamlit provides the user interface.
    - Users can enter document type, parties, terms and effective date.
    - Users can generate, preview and edit documents.
5. **Document Export**
    - TXT export
    - DOCX export with formatting, logo and table
    - PDF export with branding, headers and footers
6. **Document Formatting**
    - format_docx() formats Word documents.
    - format_pdf() creates formatted PDF documents.
    - format_html_preview() provides an HTML preview.
    - sanitize_text() cleans generated text before formatting.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	**Yes**
2	Meaningful variable and function names are used	**Yes**
3	Code includes comments / documentation where necessary	**Yes**

4	Error handling is implemented for critical operations	**Yes**
5	The application runs without critical errors	**Yes**
6	Code is committed to a version control repository	**Pending**

Additional Notes / Comments

LegalEase follows a modular architecture consisting of a **Streamlit frontend, FastAPI backend, Gemini AI integration and document-formatting modules**. The application is designed to generate, preview, edit and export legal documents in multiple formats.

The system can be extended in the future with additional legal document templates, deeper contract analysis, legal database integration, multilingual capabilities and personalized recommendations